

Student Course Completion

Data Analysis Capstone Project — Final Report

Dataset: Course_Completion_Prediction.csv — 100,000 students, 40 original fields

100,000 Total Students	49.0% Overall Completion	48 Fields Analyzed	0 critical Data Quality Issues
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This report covers the business problem, the dataset, data quality findings, business insights, and recommendations. The full code (data quality assessment, cleaning, and EDA) lives in the notebooks/ folder, and an interactive Streamlit dashboard lives in dashboard/. Every number in this report was pulled directly from running that code against the full 100,000-row dataset.

1. Business Understanding

Context and stakeholder

This dataset comes from an online learning platform offering courses across five categories: Programming, Marketing, Business, Design, and Math. The stakeholder is the platform's student success or academic operations team, the group responsible for deciding where to put outreach effort, which learners need a nudge, and where the platform is structurally losing people.

Objective and scope

The objective is to understand what actually separates students who complete a course from students who don't, using the demographic, engagement, performance, and payment signals already captured in the platform's data, and to turn that understanding into a dashboard the team can use on an ongoing basis. Scope is limited to this dataset, no external data, no production model deployment, this is an analysis and reporting deliverable.

Key business questions and what this report found

- **Which students are least likely to complete, and how early can it be spotted?** Engagement level, measurable within the first stretch of a course, is the strongest lever found. See Section 6.
- **What separates completers from non completers?** Engagement (time spent, video watched) separates the two groups more than raw performance (quiz/project scores) does.
- **Do course category or difficulty structurally affect completion?** No, not meaningfully. This was a genuine early hypothesis that the full-scale data did not support.
- **Does how a student paid relate to completion?** Yes, clearly. Free enrollments complete at a noticeably lower rate than every paid mode.
- **Is completion actually driven by progress percentage?** No. A meaningful share of completed students show low logged progress, this is explained in detail in Section 6.

2. Data Understanding

Single table, one row per student enrollment, 100,000 rows, 40 original columns spanning identifiers, demographics, course attributes, engagement behavior, performance, payment, and support/satisfaction fields, plus a binary completion target. Full field-by-field dictionary below.

Field group	Count	Examples
Identifiers	4	Student_ID, Course_ID, Course_Name (Name excluded, PII)
Demographics & access	7	Gender, Age, Education_Level, City, Device_Type
Course attributes	4	Category, Course_Level, Course_Duration_Days
Engagement behavior	11	Login_Frequency, Time_Spent_Hours, Video_Completion_Rate
Performance	6	Quiz_Score_Avg, Project_Grade, Progress_Percentage
Payment & enrollment	5	Payment_Mode, Fee_Paid, Enrollment_Date
Support & target	3	Satisfaction_Rating, Completed (target)

Full field-by-field dictionary

Field	Type	Notes
Student_ID	text	Unique key, e.g. STU175721
Name	text	Real student name, PII, excluded from all analysis
Course_ID / Course_Name	text	8 distinct courses, e.g. C101 / Python Basics
Gender	category	Male / Female / Other
Age	int	17 to 43
Education_Level	category	HighSchool, Diploma, Bachelor, Master, PhD
Employment_Status	category	Student, Employed, Self-Employed, Unemployed
City	category	15 Indian cities
Device_Type	category	Mobile, Laptop, Tablet
Internet_Connection_Quality	category	Low, Medium, High
Category	category	Programming, Marketing, Business, Design, Math
Course_Level	category	Beginner, Intermediate, Advanced
Course_Duration_Days	int	25 to 90 days
Instructor_Rating	float	4.1 to 4.7
Login_Frequency	int	Logins in the tracked window
Average_Session_Duration_Min	int	Typical session length in minutes
Video_Completion_Rate	float (%)	0 to 100 scale
Discussion_Participation	int	Count of discussion posts/replies
Time_Spent_Hours	float	Total hours spent on the course
Days_Since_Last_Login	int	Recency of activity
Notifications_Checked	int	Count
Peer_Interaction_Score	float	1.2 to 10 scale
Assignments_Submitted / Missed	int	Out of a fixed assignment count
Quiz_Attempts	int	Count
Quiz_Score_Avg	float	40.6 to 100
Project_Grade	float	20.9 to 98.8
Progress_Percentage	float (%)	24.6 to 92.4, see Section 6 for why this doesn't cleanly gate completion
Rewatch_Count	int	How often content was replayed

Field	Type	Notes
Enrollment_Date	date (parsed)	Originally stored as text in dd-mm-yyyy format
Payment_Mode	category	UPI, Free, Credit Card, NetBanking, Debit Card, Scholarship
Fee_Paid	category (Yes/No)	See the Scholarship quirk noted in Section 3
Discount_Used	category (Yes/No)	
Payment_Amount	int	Rupees
App_Usage_Percentage	int (%)	0 to 100
Reminder_Emails_Clicked	int	Count
Support_Tickets_Raised	int	0 to 4
Satisfaction_Rating	float	1.3 to 5.0
Completed	category (target)	'Completed' / 'Not Completed', 49.0% / 51.0% split

Target KPI: overall completion rate, currently 49.03%, close to perfectly balanced between Completed and Not Completed. **Analytical dimensions used throughout:** Category, Course_Level, Employment_Status, Payment_Mode, Device_Type, and the engineered engagement_tier (Low / Medium / High).

3. Data Quality Assessment

Full findings and the resolution for each issue live in **reports/quality_report.csv**, produced by **01_data_quality_assessment.ipynb**. Summary:

Issue	Scale	Resolution
Missing values	0 across all 100,000 rows	None needed
Duplicate rows / IDs	0	None needed
Out-of-range or negative values	0 across every numeric field checked	None needed
Enrollment_Date stored as text	100% of rows	Parsed to datetime in notebook 02
Fee_Paid=No but Payment_Amount>0	10,039 rows (10.0%)	Documented as a Scholarship business rule, flagged not overwritten
Completed=Yes but Progress_Percentage<50	14,342 rows (29.3% of completers)	Investigated in EDA, see Section 6
Statistical outliers (IQR)	0.1%-6.2% per field, scattered	Left as real behavior, not capped

This dataset is unusually clean, likely a synthetically generated or heavily curated dataset rather than raw production data. The two genuine issues worth attention are business-logic quirks, not data

corruption, and both are handled as explicit, documented decisions in **02_data_cleaning_preprocessing.ipynb** rather than silently patched.

6. Business Insights

Pulled directly from `03_exploratory_data_analysis.ipynb`, run against the full 100,000-row clean dataset. Ranked by gap size and actionability.

1. Engagement tier is the single strongest driver of completion found in this data

HIGH PRIORITY

Students in the High engagement tier complete at 58.2%, versus 39.7% for the Low engagement tier, an 18.5 point gap, the largest of any segment examined in this analysis. Engagement_tier is an engineered field combining login frequency, time spent, video completion, and app usage, so this isn't one behavior, it's a consistent overall pattern. This is also the most actionable finding: engagement is measurable early and the platform can intervene on it directly, unlike a fixed trait like age or education level.

2. Free enrollments complete at a meaningfully lower rate than every paid mode

HIGH PRIORITY

Free enrollments complete at 41.1%, against roughly 51-52% for UPI, Credit Card, Debit Card, and NetBanking, a 9.9 point gap. Scholarship enrollments sit in between at 45.4%. This lines up with a fairly standard pattern, financial commitment correlates with follow-through, and it gives the platform a concrete segment (Free users) to target with extra encouragement or milestone-based nudges.

3. Progress_Percentage is not a reliable signal of who will actually complete

HIGH PRIORITY

29.3% of students marked Completed had logged under 50% progress at some point, and across the full dataset, Progress_Percentage only correlates weakly with completion (0.21, the highest of any single field, but still weak on its own). Video_Completion_Rate and the engineered engagement_score rank right behind it in both the correlation analysis and a Random Forest feature importance check, two independent methods agreeing is what makes this a solid finding. Practically, this means any dashboard or early-warning system built on progress percentage alone will be unreliable, engagement-based metrics are the better signal to track.

4. Completers spend more time and watch more course video, performance scores barely differ

MEDIUM PRIORITY

Completers spend a median 3.2 hours on the course versus 2.2 hours for non completers, roughly 45% more, and their median Video_Completion_Rate is 67.7% versus 60.0%. Quiz and project scores, by contrast, are nearly identical between the two groups (74.4 vs 72.4, and 68.9 vs 67.7). This is a genuinely useful distinction: the gap between completers and non completers is about effort and engagement, not about ability. Non completers aren't performing worse when they do engage, they're engaging less.

5. Course category and difficulty level make almost no difference to completion

LOW PRIORITY

Completion rate ranges from just 48.7% to 49.2% across all five categories, and from 49.0% to 49.2% across all three difficulty levels, both essentially flat. The practical takeaway: course redesign by category is not where completion-improvement effort should go.

6. Low internet connection quality shows a small but real completion penalty

LOW PRIORITY

Students reporting Low connection quality complete at 45.8%, versus 49.6% for both Medium and High, a modest 3.8 point gap. It's real but small next to the engagement and payment findings above, worth a low-cost fix (offloadable/low-bandwidth content options) rather than a major initiative.

8. Recommendations

Each row below maps directly to one of the insights in Section 6.

Observation	Recommendation	Expected benefit
Low engagement tier completes 18.5 pts below High tier	Build an early-warning flag on engagement_score in the first 1-2 weeks; trigger outreach (email, mentor check-in) for students entering the Low tier	Closing even half this gap materially lifts overall completion
Free enrollments complete 9.9 pts below paid modes	Introduce a small refundable deposit, or milestone-based progress nudges specifically for Free-tier users	Narrows the largest payment-related gap in the data
Progress_Percentage doesn't reliably predict completion	Replace or supplement progress-percentage dashboards with engagement_score and Video_Completion_Rate as the primary at-risk indicators	More accurate at-risk flagging, fewer missed interventions
Completers spend ~45% more time and watch more video, scores are similar	Use watch-time reminders and lightweight gamification (streaks, session goals) rather than academic remediation	Targets the actual gap (effort/engagement) rather than a gap that doesn't exist (ability)
Category and Course_Level show negligible completion differences	Do not prioritize course-level redesign as a completion-improvement lever; redirect that budget toward engagement tooling	Avoids spending effort solving a problem the data doesn't support
Low internet connection quality shows a small 3.8 pt penalty	Offer a low-bandwidth or downloadable content mode for students who report Low connection quality	Removes a low-cost technical barrier for a specific, identifiable segment

Appendix: Project files

File	Contains
notebooks/01_data_quality_assessment.ipynb	Full profiling, all issues in Section 3, exports quality_report.csv
notebooks/02_data_cleaning_preprocessing.ipynb	Cleaning, feature engineering, exports clean_student_data.csv
notebooks/03_exploratory_data_analysis.ipynb	16 charts across univariate/bivariate/multivariate analysis, source of Section 6
dashboard/dashboard_app.py	Streamlit executive dashboard, run with streamlit run dashboard_app.py

File	Contains
reports/quality_report.csv	Full data quality issue log referenced in Section 3
data/processed/clean_student_data.csv	Final cleaned dataset, 100,000 rows x 48 columns